Institute of Geographical Information Systems

CS-250 - Data Structures and Algorithms LAB

Semester: Spring 2026	**Class: SCEE-IGIS - 2024**
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LAB 3: Singly Linked List

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Task 1: Create Three Nodes

```
Node* node1 = new Node(1,1.1);
Node* node2 = new Node(2,2.2);
Node* node3 = new Node(3,3.3);
```

Task 2: Add Node at End

```
void addEnd(Node* n){
    if(head == NULL){
        head = n;
        return;
    }

    Node* temp = head;
    while(temp->next != NULL)
        temp = temp->next;

    temp->next = n;
}
```

Task 3: Delete Last Node

```
Node* deleteLast(){
    if(head == NULL) return NULL;

    if(head->next == NULL){
        Node* t = head;
        head = NULL;
        return t;
    }

    Node* temp = head;
    while(temp->next->next != NULL)
        temp = temp->next;

    Node* del = temp->next;
    temp->next = NULL;
    return del;
}
```

Task 4: Count Nodes

```
int size(){
    int count = 0;
    Node* temp = head;

    while(temp != NULL){
        count++;
        temp = temp->next;
    }

    return count;
}
```

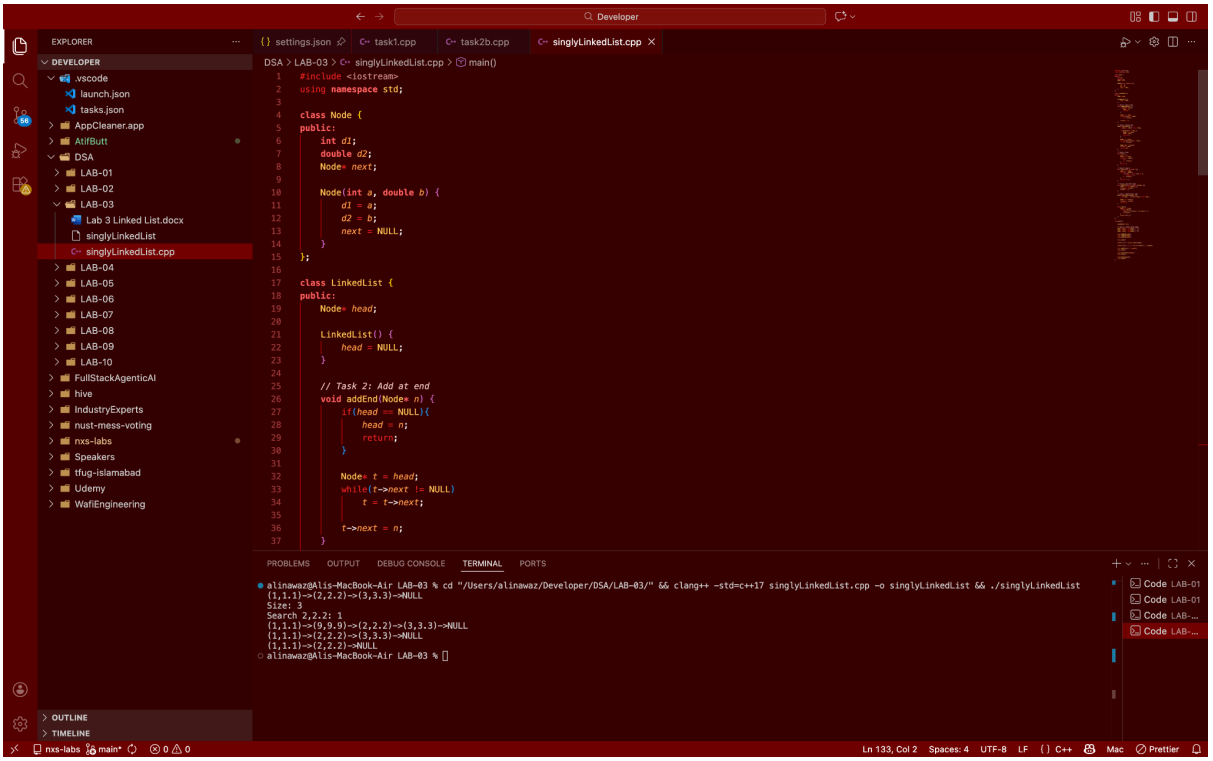
Task 5: Search Node

```
bool search(int d1,double d2){
    Node* temp = head;

    while(temp != NULL){
        if(temp->d1 == d1 && temp->d2 == d2)
            return true;
        temp = temp->next;
    }

    return false;
}
```

Task 6: Add After Given Node

```
void addAfter(int d1,double d2,Node* n){
    Node* x = new Node(d1,d2);
    x->next = n->next;
    n->next = x;
}
```

Task 7: Delete After Given Node

```
Node* deleteAfter(Node* n){
    if(n == NULL || n->next == NULL)
        return NULL;

    Node* del = n->next;
    n->next = del->next;
    return del;
}
```

Screenshot